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Import-export freight organization and optimization in the dry-port-based cross-border logistics network under the Belt and Road Initiative

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ABSTRACT

The weakness and lack of import and export logistics functions in inland regions creates a bottleneck that restricts the development of foreign economies and trade. Under the Belt and Road Initiative, the development of foreign trade demand in inland regions is stimulating the demand for inland cross-border logistics network functions. The role of dry ports in constructing cross-border logistics networks has attracted the attention of inland governments. The new dry-port-based cross-border logistics networks in Chinese inland regions began to expand rapidly, but due to the lack of operational planning and coordination, the current network efficiency is not yet well regarded. This paper focused on this new cross-border logistics network, which connects the maritime logistics network with the inland cross-border logistics network through dry ports, and studied the organizational optimization problem of inland import and export goods in different network scenarios through the application of a bi-objective mixed integer programming model using the adaptive-weight Genetic Algorithm (awGA) as a solution method. Real experiments showed the effectiveness of the method proposed by this paper. It is a new but very necessary research perspective that provides recommendations for the improvement and efficiency of inland cross-border logistics functions.

1. Introduction

The Silk Road Economic Belt and the 21st-Century Maritime Silk Road Initiatives (the Belt and Road Initiative, BRI) is a long-term major national strategy for coordinating China's all-round opening-up to the outside world, which covers 65 countries, and the economic scale of these countries accounts for approximately 29% of the entire world economy. It focuses more on cross-border and multinational cooperation between countries and Chinese regions along the BRI, offering open new opportunities for China's inland regions. Whether good participation in this regional economic cooperation initiative for related regions yields large economic benefits remains to be seen. Many inland provinces started taking an active part by striving for the new opportunities opened by the Belt and Road Initiative shortly after the initiative was launched.

Logistics plays a core role in the Belt and Road Initiative (Zhang, Ruan, Xia, & Feng, 2017). The Belt and Road Initiative prominently seeks to improve the intercommunications and innerworkings of logistics networks and optimize transportation organizations to support the free and convenient flow of international trade. Logistics plays an important role in supporting the processing of economic transactions and accelerating the development of international trade (Ducruet, Lee, & Ng, 2010; Hou & Si-Ming, 2011; Lu & Lin, 2012; Xu, Li, Shi, Zhang, & Jiang, 2015). However, inland regions of China typically have no direct international logistics functions for the export and import of commodities to and from international markets. The import-export commodities of inland regions mainly depend on maritime transport from the eastern coastal ports after a long period of land transport between inland regions and seaports which has a high cost and low efficiency. The lack of efficient import-export logistics functions is the main obstacle to the development of international trade in inland areas. Therefore, the building of international logistics channels and networks has become the primary focus of inland regions seeking to participate in the Belt and Road. In this context, dry ports with a port extension functions are developing rapidly in inland regions to overcome the drawbacks of inland regions without direct international logistics functions.

A dry port is a hinterland intermodal freight transport hub, usually providing port services such as customs declaration, inspection and quarantine and issuance of bills of lading, much like seaports do but to the exclusion of ship handling (Roso, 2007). A dry port has three key functions: port extension, cargo clustering and the assembly of multiple

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modes of transportation, which makes them very important in increasing the efficiency of inland transportation and customs clearance. Efficient inland transportation and customs clearance are essential for international logistics (Lu, 2003). At the same time, through the coordination of various modes of transportation in the logistics system, dry ports transfer cargo to rail as much as possible to reduce transport system pollution emissions, reduce logistics costs and improve the quality of transport services which yields important environmental benefits (Roso, Woxenius, & Lumsden, 2009; Samir, Nicoletta, & Francisco, 2015). So dry ports become a vital instrument for inland regions to establish their own cross-border logistics networks (Do, Nam, & Le, 2011; Monios & Wilmsmeier, 2012). Under the Belt and Road Initiative, the port extension and hub functions of dry ports, particularly the linking of land and maritime ports, have become particularly salient. A dry port is used as a starting point to connect inland regions to the Silk Road Economic Belt, and at the same time, dry ports are used to connect the inland regions to the 21st-Century Maritime Silk Road by direct cooperation with seaports. Dry ports in China are in a rapid expansion period, the number of dry ports in China has reached 136, the number of inland cross-border channels for China Railway Express (Railway Lines between China and European Countries) has reached 61, and thus the inland cross-border logistics network based on dry ports has been initially formed.

However, many problems in this rapid development of dry ports and dry-port-based cross-border logistics networks (Li, Dong, & Sun, 2015). There are a series of problems in the operation of the dry port logistics systems in China, such as the lack of overall coordination, poor organization and management, etc. As a result, essential functions of dry ports, e.g., cargo clustering and transshipment and organized junction between various modes of transportation, are not realized, further leading to the low operational efficiencies and high costs of cross-border logistics networks in inland regions of China. Therefore, it is necessary to rethink the overall network for cross-border transportation and carry out research on how to achieve efficient transportation of imported and exported goods through systemic coordination and organization. It is also important to consider how to provide decision-making reference and practical guidance to optimize the integrated cross-border logistics network system in order to address the existing problems found in the goods transport network and improve the operational efficiency of the inland cross-border logistics network so that its export-oriented functions can be truly realized.

This paper focuses on the freight flow allocation and optimization which is carried out by logistics integrators in the inland regions. The scope of the research covers the integrated cross-border logistics network, based on the hub functions of dry ports, which connects the coastal and inland areas. The aim of the research is to assist in realizing high-efficiency movements of international goods in the inland cross-border logistics network through organization of integrators and orderly processing of freight flows. In addition, the study provides some guidance for policy decisions and supportive actions for local governments to help give full support to the functions and efficiency of the inland cross-border logistics network. This is achieved through analyses of the problems found in the optimal cargo allocation schemes made by logistics integrators in different scenarios and the relevant insights that are gleaned.

The contribution of this paper lies in the cross-border logistics network based on dry ports is a new research perspective, which combines the maritime network with the land network. The focus of this paper is the realization of cross-border logistics functions in inland regions by optimizing the flow of goods in the network, which is very important but not yet studied. The dry-port-based cross-border logistics network has new structural features: (i) a broad variety of stakeholders such as shippers, logistics integrators, local governments, customs authorities, and overseas institutions; (ii) multiple types of network nodes, multiple modes of transportation, different cross-border customs clearance mechanisms, etc.; and (iii) outstanding external policy and environmental impacts. Based on these new structural features, this paper constructs a bi-objective mixed integer programming model and designs the awGA method to solve the multi-objective mathematical programming problem for different scenarios to provide decision-making recommendations for inland local governments to promote the efficient functioning of inland cross-border logistics.

2. Related work on dry-port-based logistics networks

2.1. Dry port and its role in logistics

The term “dry port” was first proposed by Munford in the 1980 transport trade journal (Munford, 1980). It worth to say that the name of the dry port has not been unified, there are various names in different institutions and occasions, such as inland terminals, inland ports, inland hubs, inland logistics centers, inland freight villages and inland clearance depots, inland container depots and intermodal freight centers, inland freight terminals and so on (Notteboom & Winkelmans, 1998; Rodrigue, Debrie, & Fremont, 2010; Roso & Lumsden, 2009). However, the study of the water - free port in the next decade did not go further. Until 2000, the inland hinterland competition is become an important aspect of seaport competition as the increasingly fierce competition among seaports, multimodal transport developed, and people begin to pay more attention on the series of environmental problems and pressure caused by container transport system, dry port started to be paid attention again and gradually become a research hot spot (Leveque & Roso, 2002).

Cullinane reviewed and analyzed the concept of dry port (Cullinane, Bergqvist, & Wilmsmeier, 2012). Although different countries, different scholars according to the different priorities given the definition conceptions, some basic characteristics and roles of dry ports are consistent, which are the essential difference to identify dry ports with other cargo terminals, cargo terminals, logistics centers. For example, dry port located in inland regions, but has the port function with convenient customs clearance, which is the most striking difference from other logistics centers. As Roso pointed out that dry port must have customs clearance function to meet the needs of customers (Roso, 2008). In addition to convenient customs clearance function like seaports, there are generally efficient logistics channels connected dry ports to seaports, such as “five-setting trains service” (the so-called five-setting trains is that railway container transport with predetermined location, time, line, shift and rate) between dry ports and seaports (Notteboom & Rodrigue, 2011). Inland import-export cargoes can finish everything except loading to the ship in dry port instead of seaports, and the inland loan-shipping agent or the operation of the shipper enterprise that needs to be completed in seaports can be realized in the local dry port (Crainic, Dell'Olmo, Ricciardi, & Sgalambro, 2015). It simplifies the operation process of goods from inland to coastal ports, improves operation efficiency, and reduces the environmental cost in transportation.

2.2. Dry-port-based logistics network planning and optimization

With the improvement of the government and industry's attention to the dry port, the system planning and logistics management of the dry-port-based logistics system have gradually become the focus areas. Overall, dry-port-based logistics system planning and optimization studies can be divided into three levels, namely, the strategic planning, tactical planning, and dry port internal operation optimization layers (Crainic et al., 2015).

The research on strategic planning and decision making in dry ports mainly focuses on the location of dry ports and the design of logistics network based on dry ports. Roso et al. pointed out the importance of rational distribution of waterless port (Roso et al., 2009). Robinson proposed a variety of spatial configuration schemes from the functional relationship between the dry port and seaport (Robinson, 2002).

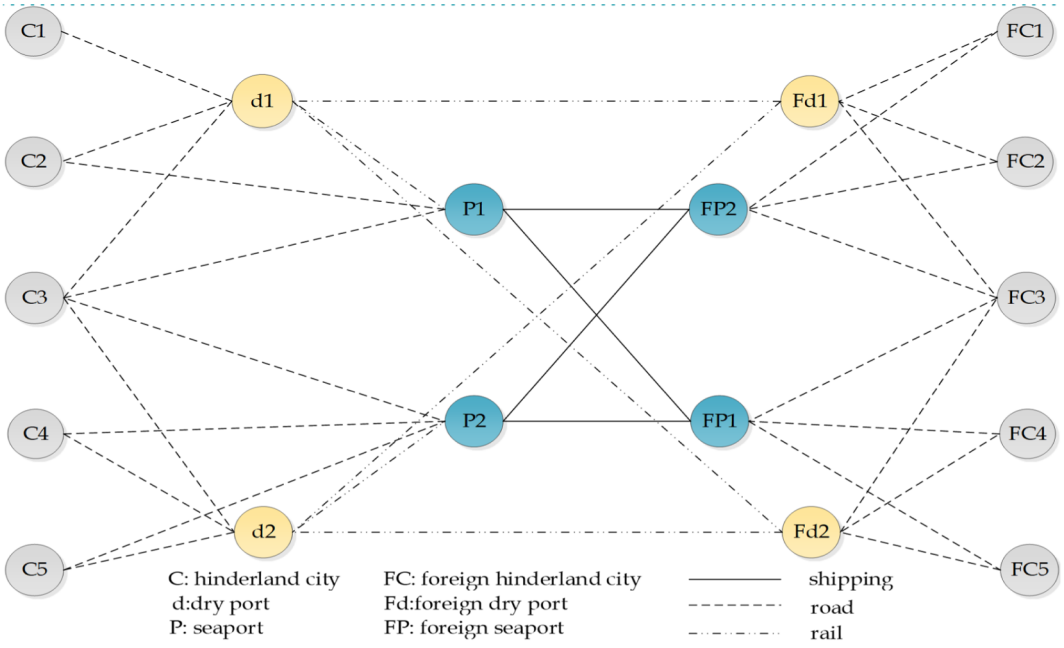


Fig. 1. The integrated cross-border logistics network based on dry ports.

Mansour used the location-allocation method to establish a new transport network architecture with a certain proportion of congestion and pollution reduction based on dry port (Mansour, Ardavan, & Robert, 2008). Henttu et al. studied the impact of the number of dry ports on economic benefits from the economic benefit (Henttu & Hilmola, 2011). Monios and Wilmsmeier reconstructed the transport chain for large ships based on the dry port (Monios & Wilmsmeier, 2012). Ambrosino et al. considered the method of joint transportation of road and railway, and studied the problem of the location of dry ports in multi-modal transport mode (Ambrosino & Sciomachen, 2014). Wang, Chen and Huang investigated the possibility of closing existing dry ports and opening new dry ports to optimize the location of dry ports (Wang & Nguyen, 2017). All the above studies are based on mathematical programming and other mathematical models. In addition, dry-port-based logistics planning involved many complicated factors such those involving policy, finance, land, and environmental issues, so there are also comprehensive evaluation and case analysis methods adopted in the studies (Chang, Lu, & Qi, 2011; Nguyen & Notteboom, 2016; Samir et al., 2015).

In terms of the dry port internal operation optimization layer, the scope of research is mainly focused on the optimization of the operation of one dry port. The research content regarding the internal operation of dry ports is similar to that of seaport operation optimization, which has produced valuable research results in the scheduling and operation optimization of human resources, vehicle resources, yard resources and so on (Francesco, Llorente, Zanda, & Zuddas, 2016; Hu, 2015; Kim, Kim, & Lee, 2016; Nguyen & Kim, 2009; Venturini, Iris, Kontovas, & Larsen, 2017). In addition, given the unique features of dry ports, there are also different research problems that arise such as in the work by Qiu et al., which considers the profit maximization and cost minimization goals of a dry port operator and studies the pricing problem of container storage by using a bi-level storage pricing model (Qiu, Lam, & Huang, 2015).

In the tactical planning of dry-port-based logistics systems, the scope of study covers dry port logistics resource allocation and scheduling, the network system operation process design and optimization, etc., and considers issues such as the organization and configuration of goods, distribution route planning and design (Crainic et al., 2015), but current research that is related to this study is scarce. Crainic et al. were the first to fill in the gaps with research of this aspect (Crainic et al., 2015). They studied optimal paths and train scheduling problems between demand/supply terminals and seaports in the dry-port-based multimodal transport system by using a mathematical model based on mixed integer programming of service network design model. However, due to the choice of the solution method, this study can only help analyze small scale problems, and the research only focused on the inland transportation between cargo supply/demand terminals and coastal ports, which is only one part of the dry-port-based logistics network. This paper focuses on the cross-border logistics network based on dry ports, which is a comprehensive network of cross-border land networks and maritime networks. The research method is based on a bi-objective mixed integer programming model and the awGA method which is more suitable for analyzing larger problems.

3. Formulation

3.1. Problem description

The inland integrated cross-border logistics network along the Belt and Road discussed in this paper, is composed based on dry ports. In the network, dry ports are connected to coastal ports, the coastal ports are further connected to overseas ports and then to overseas destinations involved in international trade. In addition, dry ports can also be connected to overseas dry ports through border ports and then to overseas destinations involved in international trade. Therefore, inland transport, maritime transport, and cross-border land transport are all included in this integrated network. Let C indicate an inland city in China, d indicate a dry port in China, P indicate a seaport in China, FC indicates an overseas destination involved in international trade along the Belt and Road, and FP indicate a port of other countries along the Belt and Road. Then the structure diagram of the integrated network could be shown as in Fig. 1.

The network provides international channels for international freight transportation between sources of supply (or destinations) in the inland of China and foreign destinations (or sources of supply). In this network, there are several ways to realize goods transport between domestic inland nodes and foreign inland nodes:

(1) Regarding domestic exports, exported goods are transported first from inland nodes in China to China's dry ports through road

transportation, then to China's seaports through rail transportation, then to foreign seaports through marine transportation, and finally to the foreign destinations where the cargo is needed through road transportation. The route is exactly the opposite for imports to China.

(2) Regarding domestic exports, exported goods are transported first from inland nodes in China to China's dry ports through road transportation, then to foreign dry ports through cross-boundary rail transportation across border ports, and finally to the foreign destinations where the cargo is needed through road transportation. The route is exactly the opposite for imports to China.

(3) Regarding domestic exports, exported goods are transported first from inland nodes in China to China's seaports through road transportation, then to foreign seaports through marine transportation, and finally to foreign destinations through road transportation. The route is exactly the opposite for imports to China.

Shippers of imported and exported goods, which are the customers of the logistics integrators, mainly take into account the following three basic aspects of freightage: quantity of goods, time of shipment, and cost of freight. Therefore, for logistics integrators, they must first meet the customers' requirements, and then meet the government's requirements in the freight market. On the premise of meeting the requirements of both the customers and government, they should try to earn an operating income through transport cost control. The focus of this study is the dry port-based cross-border logistics network, and a distinguishing feature of dry ports is that they transfer road transport to railway transport as much as possible, thus realizing a reduction in carbon emissions and environmental benefits. Therefore, they could be an effective conduit through which the government could control carbon emission rates in the logistics and transportation industry. Dry ports are effective structures on which to establish a sustainable international multimodal logistics network. In this paper, additional environmental benefits of dry ports are reflected in the government's requirements on carbon emissions in the cross-border transport market.

In summary, the problem studied in this paper is how logistics integrators can maximize their own interests through effective organization of freight flows based on the above-proposed overall structure of the inland integrated cross-border logistics network and meet their customers' different requirements in terms of freight quantity, time and cost, as well as the government's requirements on carbon emissions. The profit maximization for logistics integrators is achieved by minimizing the cost of freightage in the overall network while meeting the customers' requirements for shipment time.

3.2. Modeling

As discussed above, logistics integrators must meet the requirements of customers and the government as well as their own needs through coordinated control of freight costs and transport time and coordinated allocation of freight flows in the existing physical network. In this paper, the integrators achieve coordinated control of freight costs and transport time by making decisions on the quantity of goods and vehicles for each route, to satisfy the interests of customers, the government and themselves. Based on this, a bi-objective (freight costs and transport time) mixed integer programming model is developed in this paper and shown as follows:

(i) Variable definition

Set variables:

N the set of all nodes in the network, $N = (C, d, P, FP, Fd, FC)$, among them, C: the set of domestic inland hinterland nodes; d: the set of domestic dry port nodes; P: the set of domestic seaport nodes; FP: the set of overseas seaport nodes; Fd: the set of overseas dry port nodes; and FC: the set of overseas inland hinterland nodes.

V the set of all edges in the network. $V = V_{Cd} \cup V_{CP} \cup V_{dP} \cup V_{PFP} \cup V_{dFd} \cup V_{FPFC} \cup V_{FdFC}$.

Decision variables:

tcq_{ij} the number of containers from node i to node j;

vn_{ij} the number of vehicles dispatched from node i to node j;

Parameters:

TC_{ij} the unit transport cost per twenty-foot equivalent unit (TEU) from node i to node j;

TT_{ij} the unit transport time (hours) per TEU from node i to node j;

HC_i the unit handling cost per TEU in node i;

HT_i the unit handling time (hours) per TEU in node i;

SWC_{ij} the rail-changing cost per TEU from a domestic dry port node i to an overseas dry port node j;

SWT_{ij} the rail-changing time (hours) per TEU from a domestic dry port node i to an overseas dry port node j;

SC_i the unit storage cost per hour per TEU in node i;

ST_i the storage time per TEU in node i;

CIC_{ij} the import customs clearance cost per TEU from node i to node j;

CEC_{ij} the export customs clearance cost per TEU from node i to node j;

IIC_{ij} the import inspection and quarantine cost per TEU from node i to node j;

IEC_{ij} the export inspection and quarantine cost per TEU from node i to node j;

SQ_i the supply quantity of containers in node i;

DQ_i the demand quantity of containers in node i;

TCE_{ij} the carbon emission quantity (kg) per TEU from node i to node j;

AVN_{ij} the total number of available vehicles from node i to node j;

VQ_{ij} the transport capacity of a unit transport tool in node i;

E the maximum carbon emissions per TEU that the government or regulations allow (kg);

X_{ij} indicates whether node i is connected to node j. $X_{ij} = 1$ indicates that node i is connected to node j and there is a feasible path between them; $X_{ij} = 0$ indicates that node i is not connected to node j.

(ii) Model

$$\min f^c = \Big(\sum_{(i,j)\in V} TC_{ij} \times tcq_{ij} + \sum_{(i,j)\in V} 2 \times HC_i \times tcq_{ij} + \sum_{\substack{(i,j)\in V \\ N_i \notin FC \cup C}} SC_i \times ST_i \times tcq_{ij} + \sum_{(i,j)\in V_{dFd}} SWC_{ij} \times tcq_{ij} + \sum_{(i,j)\in V_{dFd}\cup V_{PFP}} CIC_{ij} \times tcq_{ij} + \sum_{(i,j)\in V_{dFd}\cup V_{PFP}} IIC_{ij} \times tcq_{ij} + \sum_{(i,j)\in V_{dFd}\cup V_{PFP}} CEC_{ij} \times tcq_{ij} + \sum_{(i,j)\in V_{dFd}\cup V_{PFP}} IEC_{ij} \times tcq_{ij} \Big) \div \sum_{N_i\in C} (SQ_i + DQ_i) \tag{1}$$

$$\min f^t = \Big(\sum_{(i,j)\in V} TT_{ij} \times tcq_{ij} + \sum_{(i,j)\in V} 2 \times HT_i \times tcq_{ij} + \sum_{\substack{(i,j)\in V \\ N_i \notin FC \cup C}} ST_i \times tcq_{ij} + \sum_{(i,j)\in V_{dFd}} SWT_{ij} \times tcq_{ij} \Big) \div \sum_{N_i\in C} (SQ_i + DQ_i) \tag{2}$$

s.t.

$$\sum_{(i,j)\in V} (TCE_{ij} \times tcq_{ij}) \div \sum_{i\in C} (SQ_i + DQ_i) \leqslant E \tag{3}$$

$$\sum_{(k,i)\in V} tcq_{ki} = \sum_{(i,j)\in V} tcq_{ij} \quad \forall N_i \notin FC \cup C \tag{4}$$

$$\sum_{(i,j)\in V} tcq_{ij} = SQ_i \quad \forall N_i \in FC \cup C \tag{5}$$

$$\sum_{(i,j)\in V} tcq_{ij} = DQ_j \quad \forall N_j \in FC \cup C \tag{6}$$

$$vn_{ij} \leqslant AVN_{ij} \quad \forall (i,j) \in V \tag{7}$$

$$tcq_{ij} \leqslant VQ_{ij} \times vn_{ij} \quad \forall (i,j) \in V \tag{8}$$

$$\sum_{(i,j)\in V} tcq_{ij} \leqslant CAQ_i \quad \forall n_i \in P \cup d \cup FP \cup FD \tag{9}$$

$$tcq_{ij} \in Z^+, \quad \forall (i,j) \in V \tag{10}$$

$$vn_{ij} \in Z^+, \quad \forall (i,j) \in V \tag{11}$$

where (1) and (2) are the two objective functions of the model, which are respectively the objective function for minimum logistics cost of container transportation and the objective function for minimum time of unit container transportation in the integrated cross-border logistics network. The logistics costs include road and rail transport costs, handling costs and storage costs at each node as well as rail-changing costs for cross-border rail transport, customs clearance costs and inspection and quarantine costs at ports and dry ports. It is noteworthy that the related costs incurred from changing rail (actually changing vehicles) because of different rail widths in the different countries through which cross-border trains, such as the China Railway Express, pass are called rail-changing costs. The minimum time in the objective function (2) consists of transport times, loading and unloading times and storage times. Particularly, the rail-changing times in cross-border railway transportation is included in the total time of cross-border railway transportation. The unit transport time here refers to the quotient obtained by dividing the sum of the total transport time for containers in the transport network by the total number of containers. Constraint condition (3) ensures that the carbon emission requirements of the government or other supervisory bodies are met. Constraint condition (4) describes the fact that the volume of incoming containers at each intermediate node in the network is equal to the volume of outgoing containers at the node, and this constraint is also called the freight flow balance constraint for intermediate nodes. Constraint conditions (5) and (6) indicate that both the supply and demand of containers can be satisfied. Constraint condition (7) requires that the number of transport vehicles that are available in a segment on a daily basis cannot not exceed the total number of vehicles in that segment. Constraint condition (8) ensures that the freight traffic volume of a route does not exceed the traffic capacity of the route. Constraint (9) ensures that the volume of container flows at a dry port or seaport node does not exceed the container throughput capacity of the node. Constraint conditions (10) and (11) defines the decision variables as non-negative integers.

4. Model solving

The two objectives, optimization of freight costs and optimization of transport time, of the bi-objective mixed integer programming model have obvious conflicts between each other. In addition, it is impossible to obtain the optimal values for both at the same time. According to the traditional method for solving multi-objective problems, certain weight coefficients are generally used to combine the multiple objective functions into a single objective function. The typical solution method is an evaluation function method, such as the ideal point method, linear weight method, objective programming method, objective ranking method, and objective ranking method with a relaxation factor. However, multiple objective evaluation criteria of a multi-objective problems are related to the weight coefficients which are difficult to determine, and this is a drawback for all the traditional methods for converting multi-objective problems into single-objective problems. To overcome this drawback, this paper will try to determine the reasonable solution sets for the multi-objective functions namely, the Pareto optimal solution sets for the above model.

In the studies on solving multi-objective optimization problems, it is found that genetic algorithms employ multipoint searches and have the advantage of obtaining multiple Pareto optimal solution sets through one search. In addition, the algorithm can generate a set of noninferior solutions in the solution process and thus drive the optimization towards Pareto optimal solutions. Therefore, it is an effective way to solve multi-objective optimization problems. In the process of solving multi-objective optimization problems with genetic algorithms, the most important issue is to correctly evaluate and select the Pareto optimal solutions obtained from each population generation, and retain them to the next generation. This technique avoids the omission of optimal solutions, which is caused by setting the objective weights unreasonably, thus making the search more partial to one of the objectives and unable to take into account the overall situation. The adaptive-weight Genetic Algorithm (awGA) proposed by Gen and Cheng makes use of the positive ideal points obtained from the current population in the evolutionary process and continuously refines the objective predilection according to the useful information in the current population (Gen & Cheng, 2000). The method searches for solutions through adaptive adjustment of weights to make Pareto optimal solutions approach the ideal points, thus avoiding the problems found in multi-objective optimization solution with fixed weights.

There are three basic approaches of applying genetic algorithms to a given optimization problems, namely: adapt problems to the genetic algorithms, adapt the genetic algorithms to problems, adapt both the genetic algorithms and problems (Cheng, Gen, & Tsujimura, 1999). And how to encode solutions to chromosomes is the key issue for genetic algorithms to adapt optimization management problems to the genetic algorithms (Aytug, Khouja, & Vergara, 2003; Cheng, Gen, & Tsujimura, 1996). The inland integrated cross-border logistics network studied in this paper is a bidirectional multi-objective, multi-stage and multi-mode logistics network. How to encoding chromosomes to adapt the multi-objective mathematical programming problem for this network to the awGA method is becoming a particularly key issue. The algorithm design is described as follows:

(1) Coding design

Based on the mathematical model and the variable characteristics of the model, considering the complex multi-layer structure characteristics of the integrated cross-border logistics network and multi-type mixing characteristics of cargo flow in the network, this paper adopts matrix integer encoding mode. The encoding matrix consists of five layers, as shown in Fig. 2. Taking the general network structure as an example, assuming that there is a unique OD flow from each supply node to the demand node, the OD flows starting from the domestic inland node, called forward OD flows, are coded as 1, 2, 3… *n*, and OD flows starting from foreign inland nodes, called reverse OD flows, are coded as $-(n+1)\ldots -(n+m)$. Here, *n* is the total number of forward OD flows and *m* is the total number of reverse OD flows. It is worth noting that if one supply node delivers to multiple demand nodes at the same time, the unique OD flow of each supply node can be achieved by setting up virtual supply nodes. All the OD flow coding constitutes the first layer of the encoding matrix. The second layer is the dry port coding layer through which each OD passes, the third layer is the seaport coding

OD flow direction dividing

OD flow	1	2	3	……	n	-(n+1)	-(n+2)	-(n+3)	……	-(n+m)
dry port	5	2	4	……	0	1	3	2	……	5
seaport	1	0	2	……	2	0	1	0	……	1
foreign dry port	0	3	0	……	0	2	0	1	……	0
foreign seaport	3	0	1	……	1	0	2	0	……	4

Fig. 2. The example of chromosome coding.

layer through which each OD passes, the fourth layer is the foreign dry port coding layer through which each OD passes, and the fifth layer is the foreign seaport coding layer through which each OD passes.

Specific coding rules are as follows: (a) in each layer a 0 indicates that the OD flow does not pass through the particular nodes; (b) When OD flow passes through domestic seaport nodes, it must also pass through foreign seaport nodes at the same time, and when OD flow does not pass through any seaport nodes, it must pass through foreign dry port nodes.

(2) Chromosome decoding

The decoding process is divided into the following steps. (a) Decode the transport mode of each OD flow. Taking OD flow 1 in Fig. 2 as an example, its mode of transport is from a domestic supply node through dry port 5, transport to seaport 1, then transport to foreign seaport 3, and ultimately transport to an overseas demand node. Taking OD flow $-(n+1)$ in Fig. 2 as an example, its mode of transport is from an overseas supply node through foreign dry port 2, to dry port 1, and finally transport to a domestic demand node. (b) Decode tcq_{ij}, the number of containers from node i to node j. Taking $tcq_{p_2 fp_1}$ as an example, it is the container volume from seaport 2 (p_2) to foreign seaport 1 (fp_1). In Fig. 2 $tcq_{p_2 fp_1}$ is the sum of the initial supply of OD 3, OD flow n and all OD flows passing through seaport 2 to foreign seaport 1. (c) Decode, vn_{ij} the number of vehicles dispatched from node i to node j. There is a basic rule that the next vehicle will be dispatched only when the former vehicle is full, which can be obtained from the transport capacity of a unit transport tool, VQ_{ij}, and the total transport quantity, tcq_{ij}, obtained in the previous step (c). (d) Calculate the two objective values of the corresponding chromosome $f^c(x_i)$ and $f^t(x_i)$ according to Eqs. (1) and (2).

(3) Generating the initial population generation

In this paper, the initial population generation is generated by the random generation method and is used to provide the initial points for an iterative search.

(4) Determination of the fitness function

Taking the minimization multi-objective optimization problem as an example, the process of determining the adaptive weight of awGA is as follows:

Let P represent the solution set in the feasible region and X represent one population generation of the genetic algorithm, thus X$\in$ P and X = $\{x_1, x_2, \ldots x_n\}$, where x_i, $i = 1, 2, \ldots n$ is the chromosome of an individual of X and n is the population size. Set q as the number of objective functions where k is the kth objective function, $k = 1, 2, \ldots, q$. Define the 2 vertices of each generation in the genetic algorithm:

$$Z^+ = [z_1^{\max}, z_2^{\max}, z_3^{\max}, \ldots, z_k^{\max} \ldots, z_q^{\max}] \tag{12}$$

$$Z^+ = [z_1^{\min}, z_2^{\min}, z_3^{\min}, \ldots, z_k^{\min} \ldots, z_q^{\min}] \tag{13}$$

among them, $z_k^{\max}$, $z_k^{\min}$ indicates the maximum and minimum values of the kth objective function. The definition of $z_k^{\max}$, $z_k^{\min}$ is as follows:

$$z_k^{\max} = \max\{f_k(x_i) | \, x_i \in \mathrm{X}\}, \quad k = 1, \ 2, \ \cdots, \ q \tag{14}$$

$$z_k^{\min} = \min\{f_k(x_i) | \, x_i \in X\}, \quad k = 1, \ 2, \ \cdots, \ q \tag{15}$$

Then, the sum of the weights of the objective functions corresponding to a chromosome x_i, called the adaptive weight of chromosome x_i represented as $w(x_i)$, can be obtained by the following equation:

$$w(x_i) = \sum_{k=1}^{q} \frac{z_k^{\max} - f_k(x_i)}{z_k^{\max} - z_k^{\min}} = \sum_{k=1}^{q} \frac{z_k^{\max} - f_k(x_i)}{z_k^{\max} - z_k^{\min}} \tag{16}$$

where $f_k(x_i)$ is the kth objective function of the chromosome x_i.

Accordingly, Eq. (16), the adaptive weight of the chromosome x_i in a minimization bi-objective programming model can be set as follows:

$$w(x_i) = \frac{z_1^{\max} - f_1(x_i)}{z_1^{\max} - z_1^{\min}} + \frac{z_2^{\max} - f_2(x_i)}{z_2^{\max} - z_2^{\min}} \tag{17}$$

In the adaptive weight genetic algorithms, the adaptive weight of a chromosome represents the adaptive ability of the chromosome individual in the population. The greater the adaptive weight is, the greater the probability is that the chromosome can produce offspring or be inherited into the next population generation. Therefore, according to the characteristics of the adaptive weight, this paper directly uses the adaptive weights as the Fitness Function of the chromosome. Let $f_{itness}(x_i)$ be the fitness function of the chromosome x_i. According to two objective functions Eqs. (1) and (2) of the model in Section 3.2, the fitness function of awGA for the minimization bi-objective programming model studied in this paper, $f_{itness}(x_i)$, can be expressed as:

$$f_{itness}(x_i) = z(x_i) = \frac{z_c^{\max} - f^c(x_i)}{z_c^{\max} - z_c^{\min}} + \frac{z_t^{\max} - f^t(x_i)}{z_t^{\max} - z_t^{\min}} \tag{18}$$

where $z_c^{\max}$, $z_c^{\min}$ indicates the maximum and minimum transport cost for the population generation X respectively, and $z_t^{\max}$, $z_t^{\min}$ indicates the maximum and minimum transport time for population generation X, respectively.

(5) Selection

This paper adopts the roulette method to conduct selection. Roulette selection is also called a proportional selection operator. The basic idea of the method is that the probability that an individual is selected is in direct proportion to the value of its fitness function. The selection process is as follows:

Set the population size as n, calculate the fitness function value ($f_{itness}(x_i)$) of each chromosome x_i, $i = 1, 2, 3, \ldots, n$, and then the probability (P_i) that the chromosome x_i is inherited to the next generation is written as:

$$P_i = \frac{f_{itness}(x_i)}{\sum_{i=1}^{n} f_{itness}(x_i)} \tag{19}$$

Calculate the cumulative probability of each individual. Let q_i be the cumulative probability of chromosome x_i, $i = 1, 2, 3, \ldots, n$, then q_i can be calculated as follows:

$$q_i = \sum_{j=1}^{i} f_{itness}(x_j) \tag{20}$$

Generate a uniformly distributed pseudorandom number (α) in the interval (Zhang et al., 2017). If $\alpha < q_1$, then the chromosome x_1 is selected; if $q_{i-1} < \alpha < q_i$ and $1 < i \leqslant n$, then the chromosome x_i is selected. Repeat this selection process n times.

(6) Crossover

This paper adopts the single-point crossover method. According to this method, randomly generate an integer β in $[1-n]$ as the cross point and generate two new individuals by interchanging all the genes after the cross point β of the two crossed chromosomes.

(7) Mutation

This paper adopts the uniform mutation operator. The process is described as follows: set the mutation probability as P_M, select an

individual x_i randomly from the population, uniformly and randomly generate a random number in Zhang et al. (2017) for each gene of the individual, and if $\gamma < P_M$, then assign a value within the domain, uniformly and randomly corresponding to the gene, to the gene.

(8) Elitism strategy

Studies have shown that the global optimum solution cannot be obtained by the operations of the standard genetic algorithm only, which are selection, crossover and mutation. However, this problem can be solved by the elitism strategy, whose basic idea is to carry the elite individuals, which have the best fitness values in the parental population generation, directly into the next population generation without applying crossover or mutation. This prevents excellent genes from being disrupted or lost by selection, crossover, mutation, etc. (Rudolph, 1994). In this paper, the elitism strategy is included in the operations for offspring production. That is, if the fitness value, $f_{itness}(x_i)$, of an individual x_i in the parent population is greater than or equal to the fitness value of all the individuals in the offspring population, the individual x_i will be directly included in the offspring population and then the individual with the lowest fitness value in the offspring population will be eliminated to ensure a constant population quantity.

(9) Termination criteria

This paper adopts the maximum number of iterations as the termination condition for the multi-objective genetic algorithm. In other words, the algorithm will stop once the set maximum number of iterations is achieved.

Based on the above algorithm design, the awGA for the bi-objective mixed integer programming model in this paper includes the following steps:

Step 1: Set the population size as n, crossover rate as P_c, mutation rate as P_M, maximum number of generations as $\max gen$, initial evaluation function value as $\min eval$, and initial number of iterations as 0.
Step 2: According to the chromosome coding mechanism proposed in this paper, generate the chromosomes of the initial solutions that meet the constraint conditions of the bi-objective optimization problem (coding).
Step 3: Calculate the objective function values $f^c(x_i)$ and $f^t(x_i)$ (shown in Eqs. (1) and (2)) of the bi-objective optimization problem for each chromosome (decoding).
Step 4: By using the values of the objective functions $f^c(x_i)$ and $f^t(x_i)$, generate the Pareto optimal solution.
Step 5: By using fitness function proposed in this paper shown in Eq. (18), calculate the fitness value of each chromosome and combined with the elitism strategy, carry out the selection, crossover and mutation operations for the population to obtain the next population generation.
Step 6: If $gen < \max gen$, return to Step 5. Conversely, if the termination condition is satisfied, output the Pareto optimal solution and finish the program.

5. Experiments

5.1. Experiments

In March 2015, the National Development and Reform Commission, Ministry of Foreign Affairs, and Ministry of Commerce of the People's Republic of China, with State Council authorization issued the document *Vision and Actions on Jointly Building Silk Road Economic Belt and 21st-Century Maritime Silk Road*, which gives the key inland regions in the Belt and Road, for example: "We should build Chongqing into an important pivot for developing and opening up the western region, and make Chengdu, Zhengzhou, Wuhan, Changsha, Nanchang and Hefei leading areas of opening-up in the inland regions. " in this document. Therefore, in order to be more representative, the logistics integrators are located in four inland provinces in Hubei, Hunan, Anhui and Jiangxi. The logistics integrator has five shopping centers in each of the four inland provinces. The transportation demand for each inland node is 1500 TEU. There are five dry ports in these four provinces: Wuhan, Hefei, Changsha, Shangrao and Nanchang dry ports. Five dry ports can transport cargoes to Ningbo port and Shanghai port respectively by rail or road, and then to 8 abroad seaports: port of Antwerp, Hamburg port, Le Port, Piraeus Harbor, port of Rotterdam, Venice port, port of Gdansk, port of Barcelona, and then to the hinterland by road. In addition, Hefei dry port, Wuhan dry port, and Changsha dry port, have opened three China Railway Express, which are direct to Hamburg, Merlin Pal Dube Ce, and Duisburg. The foreign hinterland nodes are composed of 11 foreign cities in which 8 foreign seaports and 3 foreign dry ports located. According to the Chinese rail mileage meter, China road trip meter, the official news data of China and Europe, and the ship information network, the data of transportation distance between each node of the network is collected.

According to the literature (Lam & Gu, 2016), and the actual data collected, various parameters, such as the transportation costs, times and speed, were set as shown in Tables 1–3. It is worth noting that the related cost parameters in Table 2 are only the costs of the cross-border inland transport related to the China Railway Express, which this paper focuses on. Regarding the characteristics of convenient customs clearance of dry ports mentioned above, for inland customers, the cost of customs clearance at local dry ports is lower than that at seaports, because the extra cost of remote operations is eliminated. Therefore, the cost of customs clearance at a dry port is different from that at a seaport. For ease of discussion, this paper divides the customs clearance cost into two categories: the cost of customs clearance at a dry port is set at 430$/TEU, while the cost of customs clearance at a seaport is 500$/TEU. The genetic algorithm parameters are set as: the population size is 100, the maximum number of iterations is 200, the crossover probability is 0.7, and the mutation probability is 0.05.

5.2. Results analysis and discussion

According to the above settings of the network and genetic algorithm parameters, a total of 50 Pareto optimal solutions were obtained by using MATLAB, and their distribution is shown in Fig. 3.

The objective function values of the 50 Pareto optimal solutions were normalized in the range [0–1], and the changing trends of cost and time in the different Pareto optimal solutions were plotted, as shown in Fig. 4.

From Figs. 3 and 4, it can be seen that the transport time and cost of inland freight in the cross-border logistics network shows opposite trends. As seen from the normalized transport time curve in Fig. 4, the rate of decrease in time decreases as the cost increases evenly from the 30th Pareto optimal solution, indicating that the increase in time efficiency achieved by an increase in cost becomes increasingly more inefficient. In other words, compared to the first 30 Pareto optimal solutions, more money is needed to achieve the same reduction in time. Therefore, this paper particularly analyzes the optimal scheme of the

Table 1
Transportation cost of per TEU under different modes of transportation (unit: $/km).

Road	Rail		Shipping
	Domestic rail	Abroad rail	
2	1.5	1.7	0.3

Table 2
Time and cost parameters in China Railway Express.

Rail-changing time (h)	Rail-changing cost ($/TEU)	Customs clearance cost ($/TEU)
0.39	590	430

Table 3
Transport speed parameters under various modes of transportation (unit: km/h).

Road	Rail	China railway express	Shipping
80	70	120	40

30th Pareto optimal solution. The minimum cost corresponding to the optimal solution is 7,579.093 USD, and the minimum time is 289.0802 h. When only the objective of minimizing costs is considered, the minimum cost is $ 6,590.528 USD and the transport time is 325.0543 h. Thus, the cost is only increased by 15% in the optimal scheme compared to minimum cost solution, in which only the cost objective is considered, while the transport time is reduced by 36 h. Therefore, the solution is called the optimal solution with a 15% relaxation of the minimum cost, and the cargo allocation scheme is shown in Table 4. Fig. 5 shows the freight flow allocation for various modes of transport and shipping routes.

It can also be seen from Fig. 3 that when only the objective of minimizing time is considered, the minimum cost is 8,238.143 USD, which is 25% higher than the minimum cost obtained when only the cost objective is considered. In other words, the integrator can additionally spend 25% of the minimum cost to achieve the shortest possible time. It also shows that through adjusting transportation costs, the relaxation space in transport time is [0–25%].

Table 4 shows a statistical summary of the freight flow allocation schemes corresponding to the scenario in which only the minimum cost is considered, the scenario in which only the minimum time is considered, and the optimal solution with a 15% increase in cost.

The following conclusions can be drawn from Table 4:

(1) When only the objective of minimum cost is considered, sea transportation is selected for all cross-border sections, and no goods are transported by China Railway Express. This result is consistent with the reality of the current inland cross-border transport system. This shows that, in the current cost structure of maritime transport and cross-border land transport, the cross-border land transport by China Railway Express is unattractive to freight flows, and the hub function of dry ports cannot be brought into play.
(2) With the pursuit of shorter transport time, cross-border land transportation begins to be selected by more freight flows, and the traffic proportion of China Railway Express increases gradually. However, even when the minimum time is set as the objective, the proportion of the cross-border railway transportation is still low, which further indicates that, when both transportation cost and transport time are considered, the current cross-border land transportation by China Railway Express is still less attractive to freight flows than the traditional sea transportation.
(3) With the increasing demand for shorter transport time, the proportion of inland road transport is declining, contrary to what one might suppose. This is because as the demand for shorter transport time increases, the China Railway Express are carrying increasingly more freight, thus diverting more inland road transport to the dry port-based rail transport. This not only reduces the environmental pollution caused by road transport but also improves the transport efficiency of cross-border logistics, showing the advantage of using dry ports to open up new routes of cross-border land transport.

According to the status quo of railway transportation in China, there is much room for improvement in both internal rail transport and China Railway Express transport. And, both internal rail transport and China Railway Express transport play a direct and key role in the development of inland cross-border transport system. If optimization measures are taken to further improve the service quality and efficiency of rail transport and reduce the unnecessary cost in railway transport, then the efficiency of the dry port-based integrated cross-border logistics

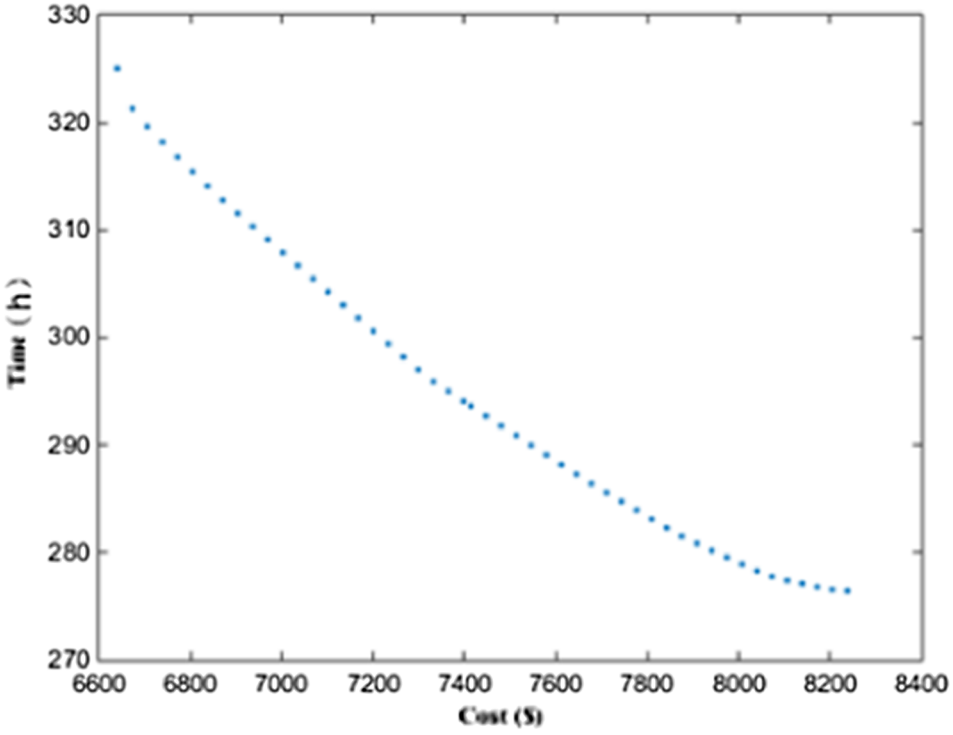


Fig. 3. Pareto optimal solution distribution.

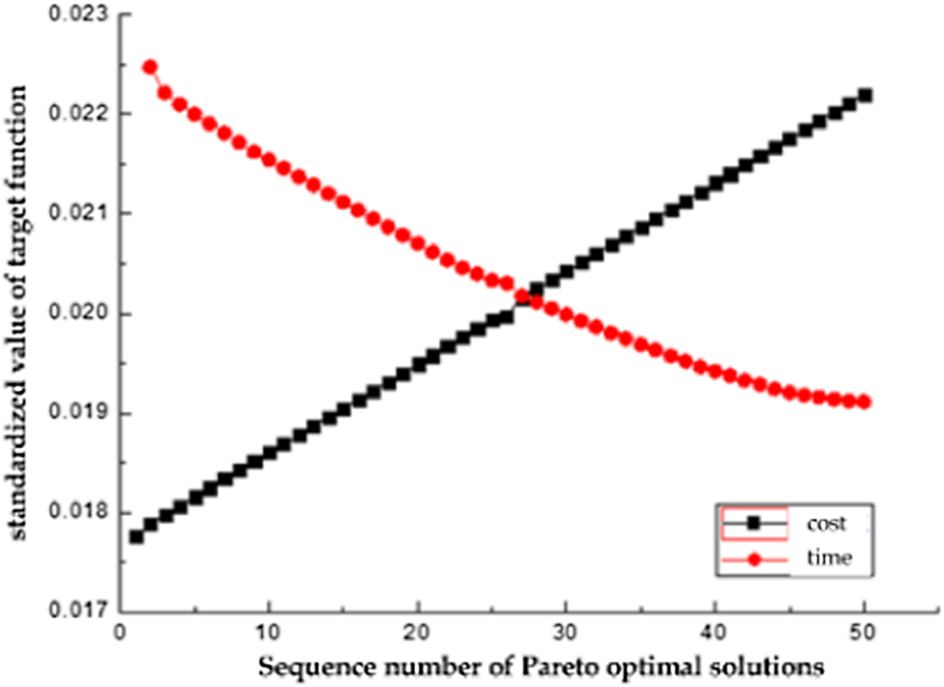


Fig. 4. The cost and time objective function values of the 50 Pareto optimal solutions.

Table 4
The optimal cargo flow configuration scheme in the network.

	Domestic inland transportation		Maritime transportation	Foreign inland transportation
	Road	Rail	Shipping	Rail
Minimum cost	48%	52%	100%	0%
Minimum time	33%	67%	64%	34%
Optimal solution with a 15% increase in cost	45%	55%	71%	29%

network will be further promoted. Furthermore, promoting the efficiency improvement and cost reduction of customs clearance, cross-border and domestic railway transportation are the places where the government is most needed to coordinate and provide actions and policy support. And actually, after the BRI was put forward, China's local governments have launched various measures, such as financial subsidies and preferential customs clearance policies, to reduce the cost of the inland cross-border railway transportation. Therefore, it is necessary to analyze the network status under different scenarios of domestic and cross-border railway transportation. Then the freight flow allocation and network efficiency of the integrated cross-border network using different optimization strategies can be analyzed. This analysis will provide some guidance for policy decisions and supportive actions for local governments to take to improve the efficiency of the inland cross-border logistics network. According to the above found in the results of the Pareto optimal analysis and the above relevant insights gained, two scenarios were designed as follow:

Scenario 1: When the cost of rail transportation is reduced by 50%, that is, the cost of rail transportation within China is reduced to 0.5 $/km and the cost of cross-border rail transportation by the China Railway Express is reduced to 0.67 $/km, the cost settings for all transportation modes are shown in Table 5.

It is noteworthy that the rail transport cost in the original network parameters is set as 1.5 $/km, because additional logistics costs caused by the poor linkages between the front end and back end of China's rail transport, such as drayage between "the gate and the station" and the related operation costs, are considered. If the additional costs caused by the poor smoothness of the railway system are ignored, then the rail transport cost can reach 0.5 $/km (Lam & Gu, 2016). The reduction of the domestic rail transport cost to 0.5 $/km can be achieved through optimization of the railway transportation system and efficient connection between dry ports and railway transportation. Therefore, Scenario 1 can be realized through optimization of the cross-border logistics system with the government's assistance or through the government's direct financial subsidies to the logistics system. At present, the latter is more common in the inland cross-border transportation system.

Using the same genetic algorithm, 63 Pareto optimal solutions were obtained. The distribution of the Pareto optimal solutions is shown in Fig. 6. Table 6 shows the optimal allocation of freight flows for different objectives in the network under the condition that the rail transport cost is reduced by 50%.

Scenario 2: When the rail transport cost, customs clearance cost and rail-changing cost are all reduced by 35% (specifically, the cost of internal rail transport is reduced to 0.97 $/km, and the cost of China Railway Express transport is reduced to 0.87 $/km) and the rail transport time is reduced by 20%, the resulting cost settings for all the transportation modes are shown in Table 7. Table 8 shows the cost and time settings for the rail changeover and customs clearance operations in the China Railway Express transport.

It is worth noting that Scenario 2 can be realized through a series of actions and measures, including process optimization of all levels of governments involved with the operational level, establishment of more effective cooperation mechanisms for customs clearance with countries along the routes, and cooperative coordination of a number of China Railway Express. Considering the current practical problems in China Railway Express transport, it would be encouraging to start with this level to improve the cross-border transportation system.

Using the same genetic algorithm, 31 Pareto optimal solutions were obtained. The distribution of the Pareto optimal solutions is shown in Fig. 7. Table 9 shows the optimal allocation of freight flows in the network for different objectives under the corresponding scenario.

Based on the adjustments to Scenario 1 and Scenario 2, the

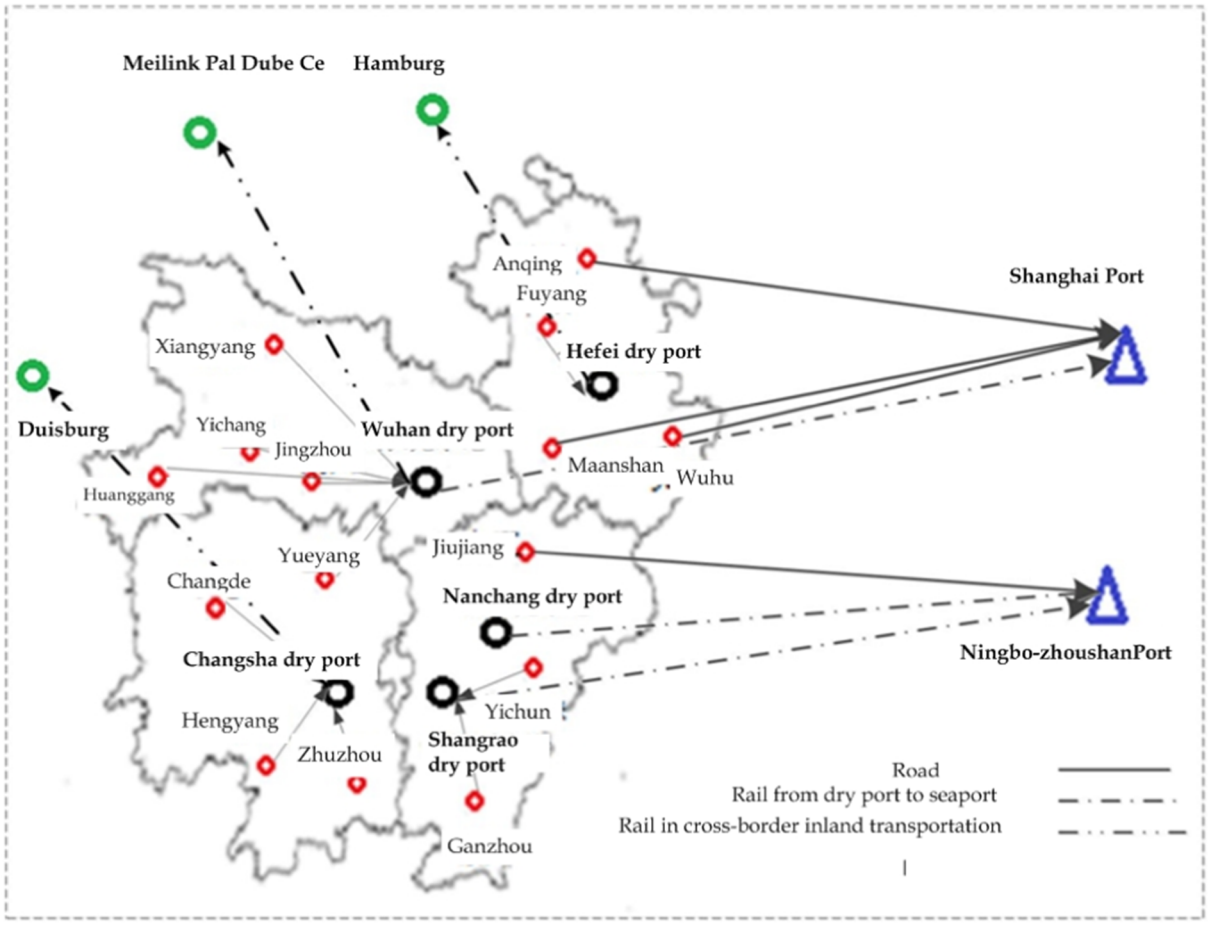


Fig. 5. Optimal flow configuration scheme with a minimum cost relaxation of 15%.

Table 5
Cost parameters of transportation modes in Scenario 1 (unit: $/km).

Road	Rail		Shipping
	Domestic rail	Abroad rail	
2	0.5	0.67	0.3

following conclusions and policy recommendations can be obtained from the distribution diagrams of the Pareto optimal solutions (Figs. 6 and 7), and Tables 8 and 9:

(1) When only the objective of minimizing cost is considered, 17% and 18% of the total freight are transported by the China Railway Express in Scenario 1 and Scenario 2 respectively. This indicates that the cross-border land transport by the China Railway Express starts to attract cross-border freight flows on the basis of the adjusted cost structure of sea transportation and cross-border land transport, which is in line with the current reality of the inland cross-border transport system. To promote the development of the China Railway Express, local governments have implemented low-price competition strategies. For example, the freight carried by China Railway Express in Wuhan, Changsha and Zhengzhou has been priced nearly equal to the price of maritime transportation as it is subsidized at a rate of 50% practically. Now, this is the only way that a certain proportion of the freight flows be attracted to the China Railway Express transport.
(2) At the initial stage of the shift of the objective function from minimizing cost to minimizing time, a small increase in cost can bring about a large reduction in time. At this point in this process, the proportion of freight flows carried by the China Railway Express rises sharply. This further shows that, in the integrated cross-border logistics network after the adjustments, the China Railway Express transport has a certain competitive advantage over traditional maritime transport when transport cost and time are both considered simultaneously.
(3) Compared to the original network, the proportion of inland road transport in the various optimal cargo distribution schemes drops to less than 30% in both of the adjusted scenarios, while the proportion of rail transport rises to 70% or even 81%. This indicates that the cargo throughput of dry ports in the integrated cross-border system is significantly increased and more road freight flows are transferred to rail transport. It thus can be seen that implementation of the above adjustments can effectively help dry ports fully utilize their functions for cargo accumulation and transshipment as well as their function as a hub for connecting various transport modes organically. This enhances the export-oriented logistics capabilities and efficiencies of the inland integrated cross-border logistics network in serving the movement of import and export goods. Thus, the adjustments can be regarded as effective measures and they indicate the direction to follow for improvement.
(4) Compared with Scenario 1, the proportion of rail transport in the optimal freight allocation scheme is higher in Scenario 2, indicating that dry ports are utilized at a higher rate in the cross-border logistics network in Scenario 2 and that the logistics efficiency of the overall network is improved. However, the effect of the adjustments in Scenario 2 is achieved in a different manner from that of Scenario 1, in which only logistics cost subsidies are used. The adjustments of the logistics cost in scenario 2 are the result of coordinated

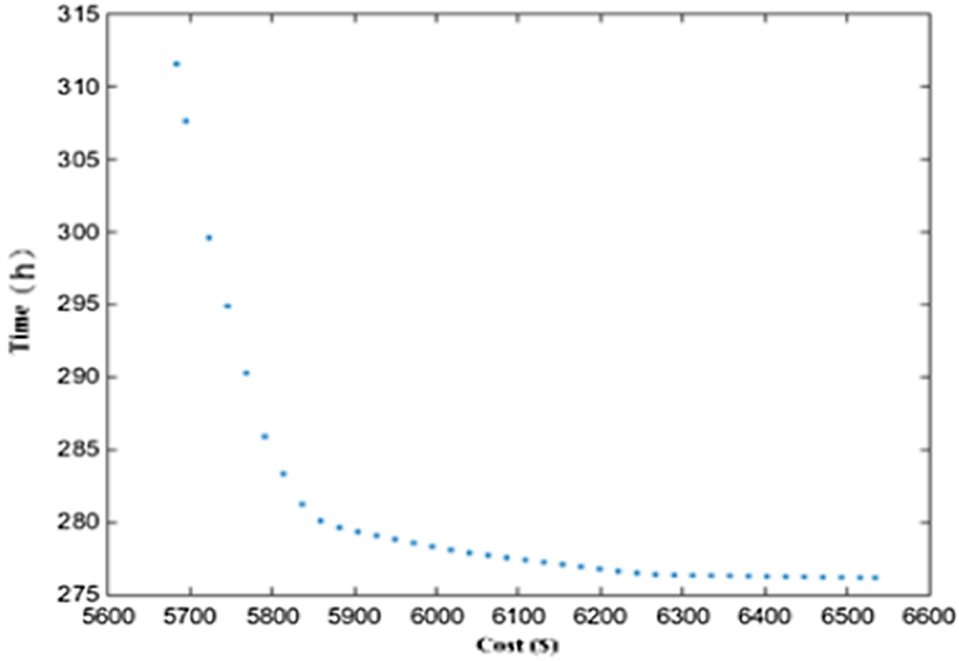


Fig. 6. Pareto optimal distribution of Scenario 1.

Table 6
The optimal cargo flow configuration scheme in Scenario 1.

	Domestic inland transportation		Maritime transportation	Foreign inland transportation
	Road	Rail	Shipping	Rail
Minimum cost	25%	75%	83%	17%
Minimum time	33%	67%	66%	34%
Optimal solution with a 15% increase in cost	30%	70%	64%	36%

Table 7
Transportation cost per TEU for different modes of transportation (unit: $/km).

Road	Rail		Shipping
	Domestic rail	Abroad rail	
2	0.97	0.87	0.3

Table 8
Time and cost parameters in the China Railway Express.

Rail-changing time (h)	Rail-changing cost ($/TEU)	Customs clearance cost ($/TEU)
0.198	383	285

optimization of all the weak links or bottlenecks in the entire cross-border logistics network. Through optimizing the weak links or bottlenecks, the logistics cost is reduced and the work efficiency is enhanced. In particular, the operation time is shortened. Therefore, even though some logistics cost parameters are higher than those in Scenario 1, the overall efficiency of the integrated cross-border logistics network is higher than that in Scenario 1 due to the shorter operation time required.

These conclusions provide recommendations for local governments to develop better policies and actions that will improve the network's operational efficiency.

6. Conclusions

The effective operation of the cross-border logistics network is the key for enabling inland regions to participate in the Belt and Road Initiative. Focusing on the dry-port-based cross-border logistics network which was rapidly established in the hinterland under the Belt and Road Initiative, this paper studied the organizational optimization problem of inland import and export freight transport in different scenarios by applying a bi-objective mixed integer programming model and an appropriate and effective solution approach called the awGA method. It is a new research perspective that is necessary, as it provides recommendations for improvements in the operation and efficiency of inland cross-border logistics network.

Based on the above explanations and from the freight flow allocations of the inland cross-border freight transport system in various scenarios, it can be seen that the hub function of dry ports, which connects land and maritime transport, is important for enhancing the efficiency of the inland integrated cross-border logistics network. Dry ports would attract freight by fully utilizing their capabilities for cargo accumulation and the transshipment and connection of various transportation modes. This is achieved through the systematic optimization of the inland and overseas transport corridors that are connected by dry ports which would establish a smooth and efficient dry port-based cross-border logistics network system. Therefore, in order to enhance the functioning of the inland cross-border logistics system, policy measures and supportive actions should be taken in this regard. The government can provide policy and operational support by promoting and coordinating mechanisms for smoother cross-border land transport cooperation. The government can also provide policy and operational support in establishing mechanisms for more efficient interfaces between dry ports and multiple modes of transport. All these government efforts could improve the efficiency of the land and cross-border land transport system and realize the full potential of the inland export-oriented logistics system and thus establish new export-oriented inland economic highlands on the Belt and Road.

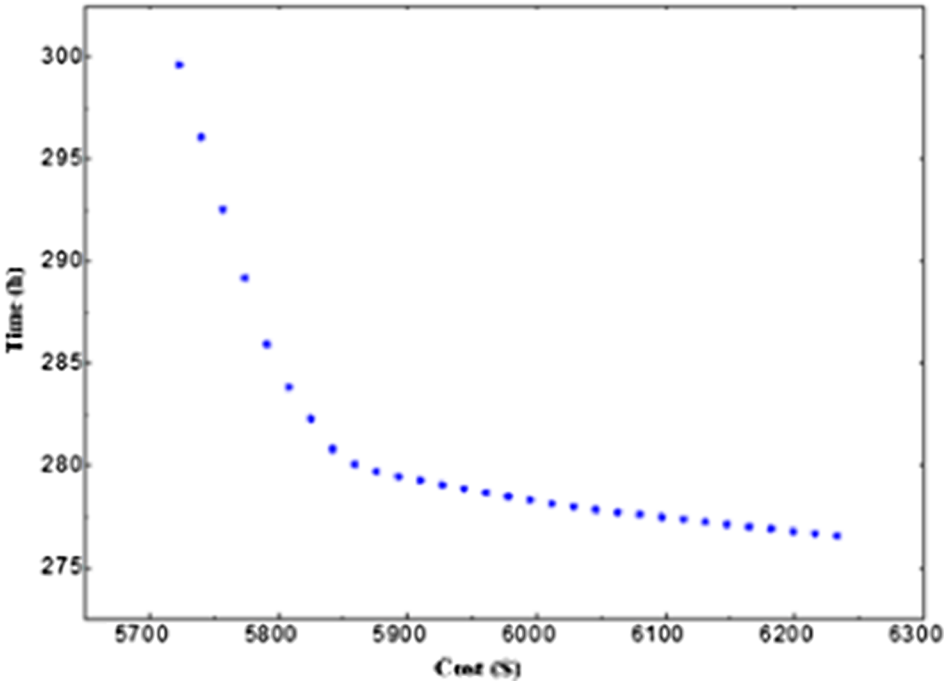


Fig. 7. Pareto optimal distribution of Scenario 2.

Table 9
The optimal cargo flow configuration scheme in Scenario 2.

	Domestic inland transportation		Maritime transportation	Foreign inland transportation
	Road	Rail	Shipping	Rail
Minimum cost	30%	70%	82%	18%
Minimum time	19%	81%	56%	44%
Optimal solution with a 15% increase in cost	20%	80%	60%	40%

Acknowledgement

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